



Louey Barbirou

Third-year Bachelor student in AI & Data · Seeking an apprenticeship in Data / Artificial Intelligence

+33 7 69 63 81 64 · louey.barbirou@ynov.com · Marseille (404 Bd Michelet, 13009), France · Driving licence B
[linkedin.com/in/louey-barbirou](https://www.linkedin.com/in/louey-barbirou) · github.com/baluva · baluva.github.io/portfolio

Profile

Third-year Artificial Intelligence & Data student at Ynov Aix-en-Provence, currently looking for an apprenticeship in data / AI (work-study rhythm: 2 weeks in the company / 1 week at school). I am doing an internship at triTon, where I develop VideoInsight (AI-powered video analysis) and work on data preparation and analysis. I enjoy hands-on projects carried end to end — from the machine learning model to the dashboard. Comfortable with Python, SQL and machine learning, I have already delivered projects in NLP, BI and predictive modelling.

Technical skills

Programming languages : Python, R, SQL, SAS, C#, Java, PHP, JavaScript, Go

Data & Machine Learning : Pandas, NumPy, scikit-learn, XGBoost, Random Forest, regression, predictive modelling, statistics, forecasting, data visualization

AI & NLP : Whisper, MarianMT, vision / LLM models (Gemini, Claude), text classification, prompt engineering

Databases : SQL, NoSQL, MongoDB, PostgreSQL, Supabase

Business Intelligence : ETL, reporting, KPIs, dashboards, Apache Superset

DevOps & Tools : Docker, Git, Linux, Windows Server, FastAPI, Agile methodology

Security : Cybersecurity, phishing detection, bug bounty / web reconnaissance

Soft skills & languages

Soft skills : Problem solving, project management, leadership, teamwork, communication

Languages : French (native), Arabic (native), English (professional — preparatory programme taught in English)

Professional experience

AI & Data Development Intern

2026 – present

triTon — Aix-en-Provence, France (4-month internship)

- Development of VideoInsight, a web application for AI-powered video analysis and translation: transcription (Whisper), translation (MarianMT), dubbing / voice cloning (XTTS), lip synchronisation and automatic scene splitting.
- Building data preparation pipelines: cleaning and structuring the audio/video datasets used to train and evaluate the models.
- Implementation of a dashboard tracking quality indicators (transcription/translation accuracy, processing time) to monitor product performance.

Stock Manager & Flow Analyst

2023

Novogel, Sfax (Tunisia)

- Analysis of sales history to forecast supply and replenishment needs.

- Monitoring of stock rotation KPIs to reduce food waste.
- Set-up of tracking tools for stock traceability and profitability.

Hospitality & catering experience

2021 - 2026

Paupiette Groupe & Junior's (Marseille), Italian Trattoria (Aix-en-Provence)

- Head chef / commis / pizza chef: menu creation, stock management and rush-hour service (up to 400 covers), teamwork.

Projects

Source code, GitHub repos and live demos for every project: see my portfolio above.

WattCast — Electricity demand forecasting in production — model forecasting French electricity demand for the next day, scored daily against RTE's official forecast: automated daily pipeline, dbt warehouse, leakage-free walk-forward backtest 2023-2026, drift monitoring and retraining. 1.90% error (MAPE) vs 1.97% for RTE.

Stack Python, LightGBM, dbt, DuckDB, GitHub Actions, FastAPI, MLflow, Evidently

SupplyPulse — Supply Chain Analytics — supply chain analysis on real data (180,000 DataCo orders): layered data warehouse (raw → staging → marts) with DuckDB, governance (data dictionary + automated quality tests in CI) and a decision-support dashboard. Key insights: 54.8% late deliveries, ~10.8% margin.

Stack Python, DuckDB, SQL, Streamlit, pandas

Prompt Booster — web application that restructures a raw prompt (role, context, output format, constraints) to improve AI answers; rule-based Python engine, web interface, deployed online.

Stack Python, Streamlit [Live demo](#)

Swarm Prediction — crop yield prediction (regression) on real FAO & World Bank data, plus a population simulation engine (SIR epidemiological model) with animated visualization.

Stack Python, pandas, NumPy, scikit-learn (Ridge, Random Forest), XGBoost

Olist — E-commerce BI dashboard — analysis of the Olist e-commerce dataset and a BI dashboard (KPIs: repeat purchase rate, delivery times) designed following data visualization principles (Few / Tufte).

Stack Python, SQL, PostgreSQL, Apache Superset, Docker

Bac Quiz — exam revision application: automatic multiple-choice question generation from PDF exam papers using multimodal (vision) AI, schema validation and a real-time backend.

Stack Python, multimodal AI (vision), Pydantic, Supabase, Expo / React Native

DBug Hunter — bug bounty assistance tool (web reconnaissance and security), deployed online.

Stack Python, Docker, Hugging Face Spaces [Live demo](#)

Phishing detection (email classification) — classification models to automatically detect malicious emails, as part of a project based on an industrial problem statement (Thales).

Stack Python, scikit-learn, NLP

Insight Hub — Student depression prediction — analysis and prediction of student depression on a 100,000-entry dataset: exploration of contributing factors (sleep, GPA, workload), interactive dashboard and a logistic regression prediction model.

Stack Python, scikit-learn, pandas, Plotly

Heart attack risk prediction — machine learning model estimating cardiac risk from clinical and lifestyle data; complete pipeline (preprocessing, scaling, training) and ROC-AUC evaluation.

Stack Python, scikit-learn, pandas

Trade With AI — crypto market prediction — crypto price prediction using time series (Prophet): trading signals (buy / sell / neutral), technical indicators and candlestick visualization exposed through an API.

Stack Python, Prophet, FastAPI

Education

Bachelor 3 — Artificial Intelligence & Data

2025 - present

Ynov, Aix-en-Provence, France

- Maths & Data Science (statistics, algebra, R, SAS, predictive modelling), advanced Python, databases (SQL, NoSQL, MongoDB).
- DevOps & management (Agile methodology), AI projects: hackathons, 48h Challenge, Ydays.

Bachelor 2 — Computer Science

2024 - 2025

Ynov, Aix-en-Provence, France

- Data & BI (advanced SQL, ETL, Python for Data, business intelligence), infrastructure & security (Docker, Windows Server, networks), C#, PHP, Java.

Bachelor 1 — IT fundamentals

2023 - 2024

Ynov, Aix-en-Provence, France

- Development & algorithms (Java, JavaScript, Go, OOP), infrastructure & systems (Linux, networks, SQL, Git).

Preparatory programme for engineering studies

2022 - 2023

South Mediterranean University, Tunisia

- Preparatory cycle taught entirely in English (North-American model).

High school diploma (Baccalaureate) — Computer Science major, with honours

2021 - 2022

Lycée 9 Avril, Sfax (Tunisia)